

Week 10 Software Lesson: Poisson Regression in R

Goals

The goal of this software lesson is to learn how to

- fit a Poisson regression model, with or without an offset term, and obtain estimates for the effect of a predictor in terms of log count/rate and in terms of rate ratios, and
- obtain model diagnostics to check assumptions and evaluate the model fit (e.g., overdispersion).

Required Packages

You will need to install the following packages in R for this software lesson:

- jtools
- emmeans
- MASS

DATASETS

Dataset #1: Lung Health Study

Let's use the Lung Health Study (LHS) again.

Recall that the dataset and data dictionary can be found on Canvas, *lhs.csv* and *lhsdoc.docx*, respectively.

```
lhs <- read.csv(file = file.choose(), header = TRUE)
```

You will use this dataset to answer the following question:

- **Model 1:** Is the rate of cigarettes smoked per day different after 3 years of a smoking intervention (with and without Atrovent) versus those who have usual care?

The variables of interest are:

- **cigsYr3:** the number of cigarettes smoked per day at year 3 of the study (we will create this variable from the *AV3CIGS* variable), and
- **alphagroup:** the intervention of either *UC* for usual care, *SI-A* for intervention plus Atrovent, and *SI-P* for intervention plus placebo.

Dataset #2: Canadian Smoking and Death Study

Another dataset used in this software lesson is the Canadian Smoking and Death Study (CSADS).

The CSADS was a survey administered to male veterans of either World War I, World War II, or the Korean War. The initial survey was administered from September 1955 through June 1956. All men (or their families) were then re-surveyed in June 1962 to determine their survival status.

The dataset and report/article of the study can be found on Canvas, titled *CanadaSmoking.csv* and *CanadianSmokingAndDeathStudy.pdf*, respectively.

```
CSADS <- read.csv(file = file.choose(), header = TRUE)
```

You will use this dataset to answer the following questions:

- **Model 2:** Does the death rate differ for those who smoke cigarettes only verses pipe/cigar only verses smoking both verses not smoking?
- **Model 3:** Does the death rate differ for those who smoke cigarettes only verses pipe/cigar only verses smoking both verses not smoking, controlling for age?

Notice that there are 4 variables and what appears to be 36 participants (i.e. 36 rows). However, each row is a summary of participants by age and smoking status. The variable `pop` is the number of veterans in each age/smoking status group, so it would be better to sum up the values in the `pop` variable to determine how many veterans were in the study.

```
sum(CSADS$pop)
```

The output tells us that there were 56,122 veterans in the study.

The variables of interest are:

- `dead`: the number who died in the 6 year period from the group,
- `pop`: the number in the group,
- `smoke`: the smoking status of the individuals in the group. There are 4 categories/levels:
 - `no`: for those who did not smoke,
 - `cigarPipeOnly`: for those who smoked either cigars or a pipe but not cigarettes,
 - `cigaretteOnly`: for those who smoked cigarettes but neither cigars nor a pipe, and
 - `cigarettePlus`: for those who smoked both cigarettes as well as at least one other form of tobacco, and
- `age`: the age group of the veteran (in 5 year increments starting with 40) (Note: will be used in Model 3 only).

Data cleaning

Creating the `cigsYr3` variable in `lhs` dataset

The `AV3CIGS` variable measures the number of cigarettes smoked per day. For most participants, this value was reported as a whole number, but approximately 50 participants reported $\frac{1}{2}$ a cigarette per day. We are going to round those participants up to 1 cigarette per day.

Because the `round()` function rounds 0.5 to the nearest even number (and therefore to 0 instead of 1), we will use the `which()` function to substitute 1 for 0.5. Because we want to preserve the original data, we will create a new variable called `cigsYr3`.

```
# creating a new variable with the same values as the old variable
lhs$cigsYr3 <- lhs$AV3CIGS

# rounding the values of 0.5 up to one in our new variable
lhs$cigsYr3[which(lhs$AV3CIGS == 0.5)] <- 1

#checking to make sure the variable was coded correctly
table(lhs$AV3CIGS)
table(lhs$cigsYr3)
```

The table for `AV3CIGS` shows that there were 49 participants who reported smoking 0.5 cigarettes per day and 18 who reported smoking 1 cigarette per day.

The table for `cigsYr3` no longer lists any participants who smoked 0.5 cigarettes per day, but there are now 67 participants who smoked 1 cigarette per day. Since $49 + 18 = 67$, our recoding worked.

MODEL FITTING

To fit a Poisson regression model, use the same `glm()` function as you saw in previous lessons, but with one exception. Use the `family=poisson(link="log")` argument to specify doing Poisson regression as opposed to some other type of generalized linear model.

Model 1: Count data with a fixed denominator

Let's see if the treatment groups (*SI-A* and *SI-P*) had any affect on the smoking rate (i.e., cigarettes per day) 3 years after the start of the study.

Note that since the variable `cigsYr3` is the number of cigarettes PER DAY, the data is already in rate form and we do not need an offset term.

```
model1 <- glm(cigsYr3 ~ relevel(factor(alphagroup), ref = "UC"),
              data = lhs, family = poisson(link = "log"))

summ(model1, confint = TRUE, ci.width = 0.95, digits = 4)
```

The model output is similar to the output for linear regression and logistic regression. (Note: To display standard error values, remove the `confint=` and `ci.width=` arguments.)

Estimate effect of predictors

Recall that in a Poisson regression model, the regression coefficients represent changes in the log count/rate scale. Also recall that you can use the results to estimate rate ratios.

Rate Ratio

- To obtain rate ratio estimates, use the code you have seen thus far, but in the `summ()` function, also include the `exp = TRUE` argument to report exponentiated coefficients (i.e., rate ratios) and confidence intervals:

```
summ(model1, confint = TRUE, ci.width = 0.95, digits = 4, exp = TRUE)
```

Rate ratio for multi-categorical predictors

For the categorical variable in the model, you may be interested in analyzing whether there are any differences in the response variable between each group after adjusting for the other predictors in the model. To estimate adjusted effects of a (multi-) categorical variable, use the `emmeans()` function from the `{emmeans}` package. By default, the function will return results on the same scale as the model. So in the case of Poisson regression, the scale is the *log* scale. To obtain estimates on the counts/rates scale, include the `type="response"` argument.

You may be asking why we use the `emmeans()` function if one can estimate the rate ratios from the model output. The model output only compares each level of a categorical variable to the reference group. For example, it doesn't give us the rate ratio comparing the *SI-P* group to the *SI-A* group. Recall that the `{emmeans}` package can provide pairwise comparisons for all groups.

```
lhs$group.f <- relevel(factor(lhs$alphagroup), ref = "UC")

model1a <- glm(cigsYr3 ~ group.f, data = lhs, family = poisson(link = "log"))

model1a.stats <- emmeans(model1a, pairwise ~ group.f, type = "response",
                        at = list(group.f = c("SI-A", "SI-P", "UC")))

model1a.stats
```

The function returns two sets of information: statistics for each group rate (under the `$emmeans` section) and statistics for differences between the groups (under the `$contrasts` section).

The output in the `$emmeans` section lists all the levels of the categorical variable in the first column. The second column (`rate`) contains the adjusted group rates for the response variable, followed by a column for its standard error (`SE`). The last two columns (`asympt.LCL` and `asympt.UCL`) include the lower and upper confidence limits, respectively.

The values presented in the `$contrasts` section include rate ratios between two groups (`ratio`), the model-estimated standard errors for the rate ratios (`SE`), degrees of freedom (`df`), *z*-statistic (`z.ratio`), and its *p*-value, after Tukey's adjustment for multiple comparisons (`p.value`).

Then, to extract confidence intervals for the rate ratios, use the `confint()` function.

```
confint(model1a.stats)
```

The function also returns two sets of information: statistics for group means (under the `$emmeans` section) and statistics for differences between the groups (under the `$contrasts` section). Except unlike before, instead of presenting *p*-values in the `$contrasts` section, 95% confidence intervals for the rate ratios (or rate ratios) are provided.

Model Evaluation

Poisson models assume that the response variable come from a Poisson distribution, which, by definition, assume that the variance (i.e., sd^2) is equal to the mean. Because of this, we need to check for overdispersion in our model.

Overdispersion happens when the observed variance is larger than the expected (or assumed) variance. To check for overdispersion, you evaluate the ratio of the residual deviance to its degrees of freedom. When this ratio is *significantly* greater than 1, then its an indication of overdispersion. Let's examine the model's residual deviance and its degrees of freedom.

```
model1$deviance  
  
model1$df.residual
```

We can see that the **Residual deviance** was 91516 and its degrees of freedom was 5463. Therefore, the ratio of these values equals 16.75. Is this value significantly larger than 1? You can test this by using the Chi-square distribution and finding the probability of observing the value of the residual deviance or larger.

```
pchisq(model1$deviance, model1$df.residual, lower.tail=FALSE)
```

Models for overdispersed data

So the model shows evidence of overdispersion. Now what? There are two ways to handle overdispersion.

Quasi-Poisson regression

One option is to use a quasi-Poisson regression model, which can be added to the `glm()` function by specifying the `family=quasipoisson(link="log")` argument.

```
model1.qp <- glm(cigsYr3 ~ group.f, data = lhs, family = quasipoisson(link = "log"))  
  
summ(model1.qp)
```

The output is the same as in the regular Poisson regression model. You will notice that while the estimates are the same between the Poisson and quasi-Poisson models, the standard errors are different. As a result, the tests may differ.

Negative binomial regression

The second option is to fit a negative binomial regression model with the `glm.nb()` function from the `{MASS}` package. The syntax is identical to the `glm()` function for the Poisson regression model, except that there is no `family=` argument.

```
model1.nb <- glm.nb(cigsYr3 ~ group.f, data = lhs)  
  
summ(model1.nb)
```

Note that there were several warning messages when creating the model. These warnings indicate that the iterative process to find the best model didn't finish before reaching the maximum number of iterations. Note that these are just warnings and a model was still found.

The output from the `summ()` function looks similar to the other two models, with the same coefficients. While the standard errors, test statistics, and *p*-values are all the same as the Poisson model, this is not always the case.

Model 2: Binary data with a variable denominator (offset term)

Recall that for fixed denominators (such as what we saw using the number of cigarettes smoked per day at the third annual visit in the LHS dataset), we don't need to specify an offset term. However, if a variable has a denominator that varies, you will need to use a Poisson regression with an offset term. So now we are going to shift to a dataset where the outcome is binary (death or not) and an offset term is needed (due to denominators varying).

For this analysis, we will be predicting the mortality rate in a 6-year period for male veterans in Canada based on the type of smoker they were. Since the data is given as number of deaths, instead of the mortality rate, we will be using the number of participants in the group as our offset term. Since the predictor of interest is the smoking type, we will also calculate the rate ratios for the `smoke` variable.

```
# Convert number of veterans per group (the offset term) to log scale
CSADS$log.pop <- log(CSADS$pop)

# Fit a Poisson regression model with log of the followup time as the offset term
model2 <- glm(dead ~ relevel(as.factor(CSADS$smoke), ref="no"),
              data = CSADS, offset = log.pop, family = poisson(link = "log"))

summ(model2, confint = TRUE, ci.width = 0.95, digits = 4)

summ(model2, confint = TRUE, ci.width = 0.95, digits = 4, exp = TRUE)
```

The output is similar to those we've seen before.

Model 3: Binary data with a variable denominator and a covariate

It seems counterintuitive that those who smoked cigarettes and other forms of tobacco would not have a significantly different mortality rate than those who don't smoke, when those who smoke only those other forms of tobacco do. Could it be that many of older people who smoked multiple forms of tobacco didn't survive to make it into the study? After all, the mortality rates are higher for the older age groups. Let's add `age` as a covariate to control for the potential age effect.

```
model3 <- glm(dead ~ relevel(as.factor(CSADS$smoke), ref="no") + age,
              data = CSADS, offset = log.pop, family = poisson(link = "log"))

summ(model3, confint = TRUE, ci.width = 0.95, digits = 4)

summ(model3, confint = TRUE, ci.width = 0.95, digits = 4, exp = TRUE)
```

The output is similar to those we've seen before.

Additional Resources

For additional R resources on Poisson regression, see:

- [Poisson Regression \(from UCLA\)](#)
- [Poisson Regression in R \(from Dataquest\)](#)
- [Poisson Regression and Model Checking \(from Duke\)](#)